Jiangyu Su (Julian)

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Education

Fudan University *Sep. 2025 – Present*
College of Integrated Circuits and Micro-Nano Electronics, Shanghai, China
Engineering (Integrated Circuits, Industry–Education Integration Track); expected graduation: 2029

GPA: 4.00/4.00 | Rank: 1st in the college

Relevant Coursework: Digital Logic Design (Honors, in progress), Linear Algebra, Computer Programming, Circuit Fundamentals, Engineering Mathematics and Probability (in progress).
Research Interests: Computer Architecture, AI Accelerators, ML Systems, Hardware–Software Co-design.

Research Experience

Causal Transformer: Learning Active Causal Strategies *2026 – Present*
Undergraduate Research Project, Fudan University

- Study Transformer policy learning and out-of-distribution generalization on unknown five-variable causal models.
- Contribute to behavior cloning from expert trajectories: five exploration steps and a final target-maximizing intervention.
- Generate synthetic data, train PyTorch models, and design controlled experiments on shortcut learning and generalization.

VULCAN-2D: Physics-Based Simulation of 2D Memristors *2026 – Present*
FDUROP Undergraduate Research Program | Research advisor: Kaichen Zhu

- Contributed to physics-based 2D memristor modeling: filament dynamics, charge transport, Joule heating, and variability.
- Refined the model against experimental I–V data; further validation pending additional measurements.

Selected Projects

AEC GPGPU: Agentic4Systems 2026 Hackathon *Jul. 2026*

- Contributed to GPGPU design for a custom 128-bit fixed-length ISA, including C reference modeling, RTL implementation, and RTL/C-model co-verification.
- Performed synthesis and PPA optimization using the ASAP7 flow; won third prize, ranking 4th among 22 teams with 48/50 on functional tests.

Tank Trouble: FPGA-Based Game System *Fall 2026*
Digital Logic Design (Honors) | Verilog, Vivado, Basys3

- Designed modular RTL integrating PS/2 keyboard input, 640×480 VGA output, ROM-based sprites with 32 orientations, and fixed-point tank movement and rotation.
- Implemented configurable maze rendering with three map sizes and wall-data storage; verified RTL through simulation, synthesis, and Basys3 board-level testing.

Raspberry Pi Smart Car: Vision and Speech Control *Fall 2025*
Introduction to Embodied Intelligence | Team Project | GitHub

- Co-developed a PC-side perception and control system for a Raspberry Pi car, integrating YOLOv5 object detection, OpenCV tracking, and PID steering via HTTP and live camera streaming.
- Integrated Silero VAD and SenseVoice/FunASR for speech commands and mode switching; explored reduced detection frequency and frame-to-frame tracking to improve real-time performance.

Honors & Awards

National Scholarship, China *2026*
Third Prize, Agentic4Systems 2026 Hackathon *2026*
Third Prize, Chinese Mathematics Competition (Non-Math A, Shanghai prelim.) *2025*

Technical Skills

Languages: C, Python, Verilog
Frameworks & EDA: PyTorch, OpenCV, Vivado
Development Tools: Linux, Bash, Git, Docker, SSH, tmux, LaTeX